

Improving Bug Report Quality in a Small Developer Team

An Action Research Study in the Colorife Mobile Application

Abstract

Bug reports are a central communication artifact in software development teams. In small teams and side projects, bugs are often reported informally through messaging applications, which can result in unclear and incomplete descriptions. This report presents an **action research study** conducted within a small developer team working on the **Colorife** mobile application. Initially, bugs were reported through a shared WhatsApp group and stored as informal text messages. For this assignment, these messages were collected and transformed into a structured dataset. A structured bug report template was then introduced, and subsequent bugs were reported using the template. A qualitative comparison of bug reports before and after the intervention shows clear improvements in clarity, completeness, and reproducibility. The study demonstrates how simple process changes can improve software engineering practices in small development teams.

1. Introduction

Bug reporting enables developers to identify, understand, and fix software defects. High-quality bug reports typically include a clear problem description, steps to reproduce, expected and actual behavior, and environment information. When such information is missing, developers must seek clarification, which slows down development and increases communication overhead.

In small developer teams and side projects, bug reporting is often informal. Messaging tools such as WhatsApp are commonly used because they are quick and convenient. However, this informality often leads to vague and inconsistent bug reports.

Colorife is a mobile application developed by a small team as a side project. During development, bugs were usually reported in a shared WhatsApp group. This report investigates whether introducing a structured bug report template improves bug report quality in this context.

2. Research Question and Objectives

Research Question

How does introducing a structured bug report template affect the quality of bug reports in a small developer side project?

Objectives

- To analyze informal bug reports shared via WhatsApp
 - To introduce a structured bug report template
 - To compare bug reports before and after the intervention
 - To evaluate the impact of the template on bug report quality
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3. Research Method

This study follows an **Action Research** methodology.

Action research is appropriate because:

- The researchers are also members of the development team
- The goal is to improve an existing practice
- The method emphasizes intervention, observation, and reflection

The study follows one action research cycle:

1. Observation of the existing bug reporting practice
 2. Introduction of a structured bug report template
 3. Observation of changes in bug report quality
 4. Reflection on outcomes
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4. Context and Study Design

Project Context

- **Project Name:** Colorife
- **Type:** Mobile application (React Native)
- **Team Size:** 5 developers (Qasim, Ibad, Arsalan, Ali, Saeed)
- **Development Style:** Side project
- **Initial Bug Reporting Tool:** WhatsApp group chat

Existing Bug Reporting Practice

Before the intervention, bugs were reported as short free-text messages in a WhatsApp group. These reports were typically informal and lacked consistent structure.

For this study:

- WhatsApp bug-report messages were collected
 - Messages were converted into a structured **JSON dataset**
 - This dataset represents the **before-intervention** state
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5. Intervention: Bug Report Template

After reviewing the informal WhatsApp bug reports, the team introduced a **structured bug report template** to standardize bug reporting.

Bug Report Template

Each bug report included the following fields:

- Bug Title
- Reported By
- Date & Time
- Environment (platform, device, OS, app version)
- Preconditions
- Steps to Reproduce
- Expected Behavior
- Actual Behavior
- Severity
- Attachments (optional)

The template was shared with the team and used for all subsequent bug reports.

6. Data Collection

Two datasets were used in this study.

6.1 Bug Reports Before the Intervention

- Bugs reported informally via WhatsApp
- Short, unstructured free-text messages
- Converted into JSON format for analysis

6.2 Bug Reports After the Intervention

- Bugs reported using the structured template
- Stored directly in JSON format
- Followed a consistent and complete structure

Both datasets contained comparable types of bugs related to authentication, navigation, movie recommendations, and health-tracking features.

7. Data Analysis

A **qualitative comparative analysis** was conducted to examine differences between bug reports before and after the intervention.

The analysis focused on:

- **Clarity** – how clearly the issue was described
- **Completeness** – whether essential information was included
- **Reproducibility** – whether another developer could reproduce the bug

The analysis was descriptive and reflective, consistent with action research principles.

8. Results

Bug Reports Before Introducing the Template

- Reports were short and informal
- Steps to reproduce were often missing
- Expected and actual behavior were unclear or combined
- Environment details were rarely included
- Follow-up questions were frequently required

Bug Reports After Introducing the Template

- Reports followed a consistent structure
- Steps to reproduce were clearly documented
- Expected and actual behavior were explicitly separated
- Environment information was regularly included
- Bugs were easier to understand and reproduce

Overall, the structured template led to **clearer, more complete, and more actionable bug reports**.

9. Discussion

The findings indicate that informal bug reporting through chat tools can negatively affect communication efficiency, even in small teams. Introducing a simple template encouraged developers to reflect more carefully on the bugs they reported and improved shared understanding within the team.

This study shows that meaningful improvements in software engineering practice can be achieved through lightweight, low-effort process changes.

10. Limitations and Threats to Validity

- The study focuses on a single project
- The team size is small
- Results are based on qualitative assessment
- Findings may not generalize to all development contexts

Despite these limitations, the study provides practical insights relevant to small development teams and side projects.

11. Conclusion

This action research study examined how introducing a structured bug report template affects bug report quality in the **Colorife** mobile application. By transforming informal WhatsApp bug reports into structured reports and adopting a template-based approach, the development team significantly improved clarity, completeness, and reproducibility of bug reports.

The study demonstrates how evidence can be created through reflective practice and highlights the value of simple process improvements in real software development settings.